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AI-Enabled Online Adaptive Learning Platform and Learner's Performance: A Review of Literature

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Abstract: The development of AI-enabled technologies and its wide uses provided new edge in the field of education and redefined the process of delivery of education. The deployment of Artificial Intelligence in education presented various teaching opportunities, learning potentials and challenges during the practice of modern pedagogies and capable to personalise the learning experiences of each learner. Online-learning-platforms able to provide the adaptive learning environment handle huge learners' data to deeply intercept the unique learning need of the student. The review of literature is trying to showcase an overview of research executed on the impact of AI-enabled online adaptive learning platforms on student's performance.

Keywords: AI-Enabled online adaptive learning, Learner's Performance, Personalised Learning, Artificial Intelligence (AI), Machine Learning (ML)

JEL Classification Number: I21, I23

1. Introduction

An AI-enabled online adaptive learning platform is an educational tool that leverages artificial intelligence (AI) technology to personalize the learning experience for individual students. Chaiyarak (2021) explains that online learning materials, combined with intelligent algorithms, deliver customized instruction tailored to the unique needs and abilities of each learner. According to Kabudi (2021), the platform typically employs data analytics and machine learning techniques to gather information about the student's progress, learning preferences, and areas of difficulty. As Das (2023) highlights, the data is utilized to create personalized recommendations, adjust the content and difficulty of the learning materials, and offer timely feedback and support to the student. Ezzaim (2022) emphasis the goal of an AI-enabled adaptive learning platform is to optimize the learning process by tailoring it to the specific needs and learning styles of each student. By adapting the content, pace, and difficulty of instruction, the platform aims to enhance engagement, motivation, and overall learning outcomes. Features of AI-enabled adaptive learning platforms may include:

a) Personalized Learning Paths: Peng (2019) considered that the platform creates customized learning paths for each student, identifying their strengths and weaknesses and suggesting appropriate learning activities and resources.

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b) Adaptive Content: Ezzaim (2022) believes that platform adjusts the difficulty level, pace, and sequence of learning materials based on the student's performance and progress, ensuring an optimal level of challenge.

c) Intelligent Assessments: According to Reichardt (2021) the platform employs AI algorithms to assess the student's knowledge and skills, providing real-time feedback and adapting the assessment based on their responses.

d) Data Analytics: The platform collects and analyses data on student performance, engagement, and learning behaviours to gain insights and generate actionable recommendations for teachers and administrators according to Minn and Sein (2022).

e) Virtual Tutoring: Some platforms incorporate virtual tutoring capabilities, using AI-powered chatbots or virtual assistants to provide immediate assistance and support to students as per the research of Zliobaite (2012).

f) Gamification and Interactive Elements: To enhance student engagement, adaptive learning platforms often incorporate gamified elements, interactive exercises, simulations, and multimedia contents are supported by Tenório (2020).

g) Integration with Learning Management Systems: These platforms can be integrated with existing learning management systems (LMS), allowing seamless integration into the educational institution's infrastructure replicated in the research work of Ishak (2016).

h) Enhanced Engagement and Motivation: By tailoring the learning experience to the individual learner, adaptive platforms can increase engagement and motivation. Learners are more likely to stay engaged when the content is relevant, challenging, and presented in a format that suits their learning style (Ishak, 2020).

i) Personalized Feedback and Support: Adaptive learning platforms can provide immediate and personalized feedback to learners, guiding them towards the correct answers or offering explanations and additional resources. This helps to reinforce learning and address misconceptions as explored by Khosravi (2020).

j) Continuous Progress Monitoring: Alam (2022) identified various AI algorithms continuously monitor and track the learner's progress, providing insights on their growth and areas for improvement. Educators can use this information to intervene and provide additional support when necessary.

k) Scalability and Accessibility: AI-enabled adaptive learning platforms can cater to many learners simultaneously, making education more scalable and accessible. According to Cai (2018) learners can access the platform from anywhere with an internet connection, enabling remote and flexible learning opportunities.

Overall, AI-enabled adaptive learning platforms have the potential to revolutionize education by providing personalized, data-driven instruction and support, improving student outcomes, and enabling more efficient use of educational resources.

1.1. Online Learning and Role of Technology in Process of Learning

Online learning or e-learning platforms are directly associated to the extensive use of digital technology to deliver educational content and facilitate learning remotely supported by Zliobaite (2012). It has gained significant popularity and recognition in recent years, offering flexibility, accessibility, and a wide range of educational opportunities to learners of all ages and backgrounds. The modern IT (Information Technology) ecosystem plays a crucial role in enabling and enhancing the online learning experience in several ways:

a) **Content Delivery:** Technology provides a platform for delivering educational content in various formats such as text, audio, video, interactive modules, and multimedia presentations. Digel (2023) elaborated that online learning platforms allow learners to access a vast array of resources, including e-books, lecture videos, simulations, and virtual laboratories.

b) **Accessibility and Flexibility:** Online learning breaks down the barriers of time and place, allowing learners to access educational materials and participate in courses from anywhere at any time. This flexibility is particularly beneficial for individuals with busy schedules, working professionals, or those who are unable to attend traditional classroom settings.

c) **Interactivity and Engagement:** Technology enables interactive learning experiences through discussion forums, chat rooms, video conferencing, and virtual collaboration tools. These features promote active engagement, peer interaction, and knowledge sharing among learners, enhancing the overall learning experience considered by Almohammadi (2013).

d) **Personalization and Adaptive Learning:** Technology facilitates personalized learning by leveraging data analytics and AI algorithms to tailor instruction to individual learners' needs. Adaptive learning platforms assess learners' performance and provide customized content, feedback, and recommendations based on their progress, strengths, and weaknesses explored by Tan (2022).

e) **Assessment and Feedback:** Online learning platforms offer various assessment methods, including quizzes, assignments, and online exams, with automated grading and immediate feedback. Jing (2023) intercepted that technology allows for efficient tracking of learner progress and performance, enabling timely interventions and personalized support.

f) **Communication and Collaboration:** Technology tools like email, discussion boards, and video conferencing enable effective communication and collaboration among learners, instructors, and peers. Learners can engage in group projects, share ideas, and receive guidance from instructors and classmates.

g) **Resource Management:** Technology streamlines administrative tasks in online learning, such as course registration, grading, and record-keeping. Learning management systems (LMS) provide centralized platforms for course organization, content management, and communication, simplifying the management of online learning environments.

h) **Continuous Learning and Professional Development:** Online learning platforms offer opportunities for continuous learning and professional development, allowing individuals to acquire new skills, update their knowledge, and stay relevant in a rapidly changing world. Massive Open Online Courses (MOOCs) and online certification programs provide accessible and affordable options for lifelong learning.

While technology is a valuable enabler of online learning, it is important to recognize that effective implementation requires thoughtful instructional design, pedagogical strategies, and support systems. The role of educators in designing and facilitating meaningful online learning experiences remains essential in leveraging the full potential of technology for education.

1.2. Impact of AI-enabled adaptive learning platform on learner's performance

AI-enabled adaptive learning platforms can have a significant impact on learners' performance measured by Kabudi (2021). Here are some ways in which these platforms can positively influence learner outcomes:

a) Personalized Instruction: Adaptive learning platforms provide personalized instruction tailored to each learner's unique needs, strengths, and weaknesses. By adapting the content, pace, and difficulty level of instruction, these platforms ensure that learners receive targeted and appropriate learning materials. This personalized approach can enhance understanding, engagement, and retention of knowledge, ultimately leading to improved performance marked in the research of Ezzaim (2022).

b) **Targeted Remediation:** Adaptive platforms identify areas of weakness or misconceptions in learners' understanding through continuous assessment and data analysis. They provide targeted remediation and additional practice in these areas, helping learners overcome challenges and reinforce their understanding.

Kabudi (2021) articulated that those targeted support can boost performance and bridge learning gaps.

c) **Optimal Challenge Level:** Adaptive learning platforms dynamically adjust the difficulty level of content based on learners' progress. They ensure that learners are appropriately challenged, neither overwhelmed nor bored. By maintaining an optimal challenge level, these platforms keep learners motivated and engaged, which can positively impact their performance.

d) **Real-Time Feedback:** AI-enabled adaptive platforms offer real-time feedback to learners. They provide immediate information on correctness and offer explanations or hints when learners make mistakes. This timely feedback helps learners understand their errors, correct misconceptions, and make improvements. It enables learners to address their weaknesses promptly, leading to enhanced performance considered by the Minn (2022).

e) **Mastery Learning:** Adaptive platforms often employ mastery learning principles, requiring learners to demonstrate proficiency in one topic or skill before progressing to the next. Learners must achieve mastery before moving forward, ensuring a solid foundation of knowledge. This approach can significantly impact performance by promoting depth of understanding and mastery of concepts also reflected in the research of Jing (2023).

f) **Data-Driven Insights:** Adaptive learning platforms gather and analyse data on learners' performance, behaviours, and preferences. Educators can utilize these insights to gain a deeper understanding of learners' strengths, weaknesses, and learning patterns. This information allows educators to make informed instructional decisions, provide targeted interventions, and customize learning experiences to optimize learner performance supported by Xu (2022).

g) **Continuous Progress Monitoring:** Adaptive platforms continuously monitor learners' progress and provide ongoing feedback. Educators can access this information and intervene when learners are struggling or falling behind. Cai (2018) established that the timely interventions and support can prevent learners from developing persistent gaps in knowledge and skills, leading to improved performance.

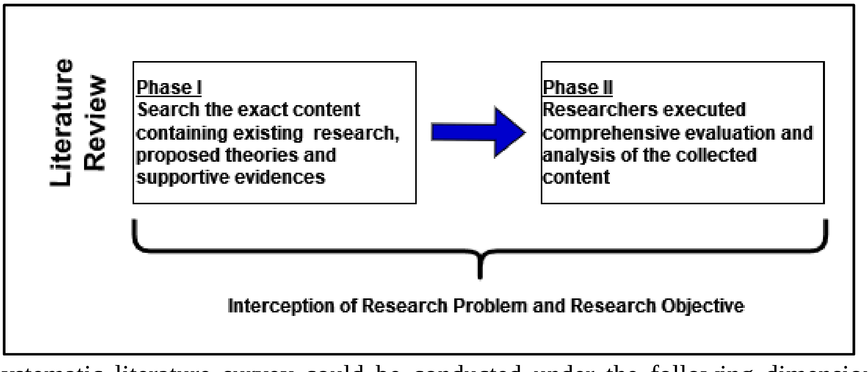
h) **Self-Paced Learning:** Adaptive platforms often offer self-paced learning options, allowing learners to progress at their own speed. Learners can take the time they need to master concepts, revisit challenging topics, or accelerate through familiar material. This flexibility supports individual learning preferences and can positively impact performance by enabling learners to learn at their optimal pace, this is also realised by Ouyang (2022).

Overall, AI-enabled adaptive learning platforms can significantly impact learners' performance by providing personalized instruction, targeted support, optimal challenge, timely feedback, and data-driven insights. These platforms have the potential to enhance understanding, motivation, and engagement, leading to improved learning outcomes and academic achievement.

2. Methodology for the proposed Literature Review

This comprehensive literature review provides an overview of the current state of AI-enabled adaptive learning platforms. The review explores the definition of adaptive learning, the role of artificial intelligence in enabling adaptivity, and the benefits and challenges associated with these platforms, this is also supported by Jing (2023). Furthermore, it highlights recent research studies, advancements, and trends in the field of AI-enabled adaptive learning platforms.

Figure 1: Interception of Research Problem and Research Objective



The systematic literature survey could be conducted under the following dimension to collect more useful information: -

- A systematic mapping of all available online adaptive learning platforms
- Advantages of AI-Enabled Online Adaptive Learning Platforms
- Future Potentials and present challenges of AI-Enabled adaptive learning platforms

There are various published papers on Adaptive Learning and lot of researchers influenced by this area of research. There are many aspects of Adaptive Learning that are worth exploring in depth. For instance, one could delve into the various ways in which Adaptive Learning can be implemented in different educational settings. Additionally, it would be interesting to examine the impact of Adaptive Learning on student outcomes, such as retention rates, test scores, and overall academic achievement. Another area of inquiry could be the potential benefits of Adaptive Learning for students with different learning styles or special needs. Furthermore, there are many challenges that must be overcome to successfully implement Adaptive Learning, such as ensuring that the technology is accessible to all students and that it is properly integrated into the curriculum. Overall, Adaptive Learning is a rich and complex topic that offers many opportunities for research, exploration, and innovation mentioned in the work of Peng (2019). The number of papers published by Google Scholar, IEEE Xplore and ACM on adaptive learning has been steadily increasing in recent years, as the technology has become more widely adopted for research and application.

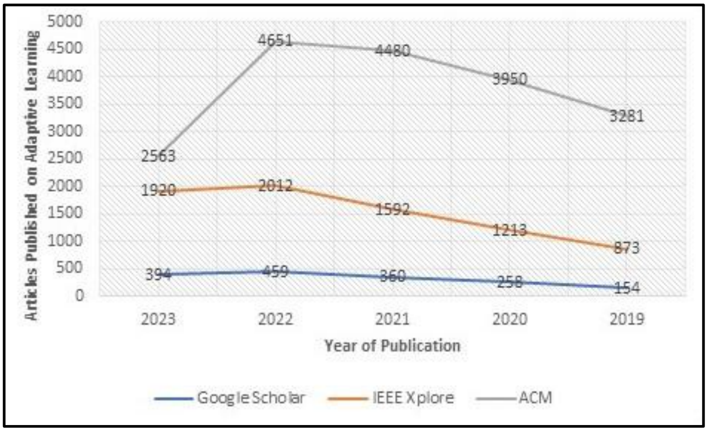
In 2023 the 5 countries that have published the most papers on adaptive learning in IEEE Xplore is as follows:

- United States (392 papers)
- China (231 papers)
- India (134 papers)
- United Kingdom (102 papers)
- Germany (92 papers)

The topics that have been mostly covered in papers on adaptive learning in IEEE Xplore, ACM and Google Scholar are:

- Adaptive learning systems
- Adaptive learning algorithms
- Adaptive learning content
- Adaptive learning evaluation
- Adaptive learning for specific domains (e.g., healthcare, education, etc.)

Figure 2: Publication Year Wise



The proposed study is conducted in systematic approach PRISMA technique known as Preferred Reporting Items for Systematic Reviews and Meta-Analyses. O'Dea, (2021) considered that it is widely used in research for the structured reporting and meta-analysis of available contents.

Advantages of adopt the PRISMA approach

- Showcase the quality of the review for AI enabled adaptive learning and learner's performances
- Make capable to researchers to assess strengths and weaknesses
- Iterative approach of review methods
- Structure approach of review under the various phases of PRISMA

The different phases of PRISMA technique could be involve in customised way for the study of AI enabled adaptive learning platforms and impact in learners' performances. The suggested phases are as follows:

Phase 1: Clearly architect the research questions and research objectives associated with on AI-Enabled adaptive learning platforms.

Phase 2: Align the PRISMA framework with proposed research objectives and available literatures on AI enabled adaptive learning online platforms.

Phase 3: Execute comprehensive and systematic search research databases to trace the relevant articles.

Phase 4: Framing of inclusion and exclusion criteria to select the valid research papers related with the concept of AI-enabled adaptive learning.

Phase 5: Establishment of Parameters for the study such as research methodologies and designs, data collection methods, different demographics used for research, use of various AI algorithms and interactive online platforms.

Phase 6: Systematic approach to extract data and collect the useful information from the selected research papers or proceedings about AI-enabled adaptive techniques.

Figure 3: PRISMA

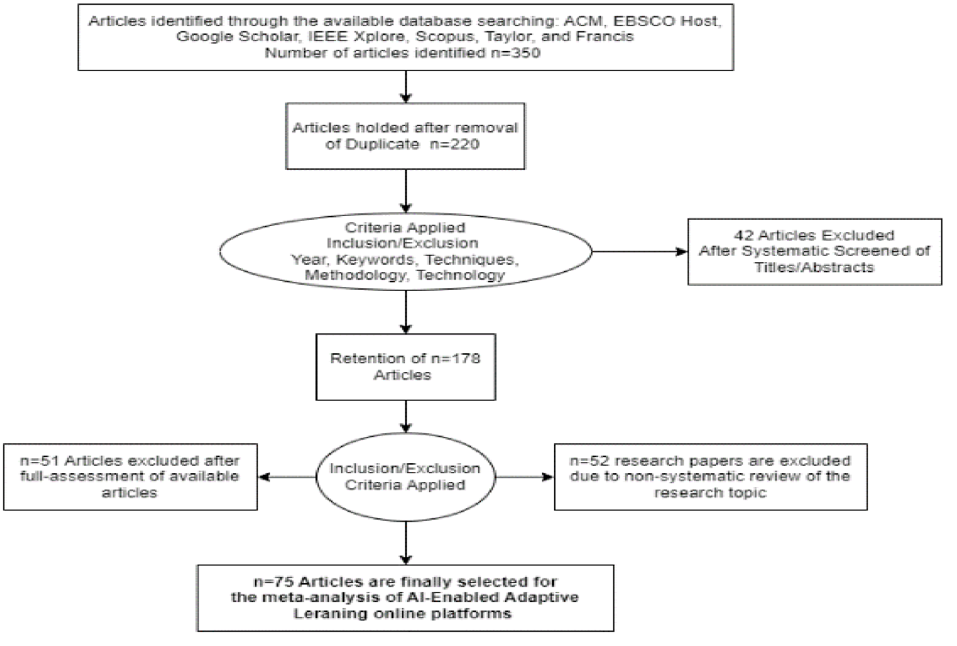


Table 1: Keywords Search

Types of Manuscripts / Articles adopted	**Search Key-words**	**Research-Databases**
Research Papers or articles from peer reviewed reputed journals Proceedings from International or National Conferences Thesis Book Chapters	Online/e-learning Platforms Adaptive Learning AI-Enabled Educational Technologies Intelligent Tutoring System Learning Management System Learners' Performances Learners' Engagement AI-Enabled Adaptive Learning Platforms	• Google Scholar • Science Direct • Inder science • Emarald • Taylor & Francis • Springer • IEEE Xplore • Ebsco

To refine the search through the keywords, Boolean operators are used to connect the concept and keywords. It helps to search useful information by keywords in search queries and to explore the suitable research papers. Using Boolean operators such as "AND", "OR", and "NOT" in a search engine will allow to construct a more precise search query and get more relevant search results.

Table 2: Boolean Operator

Boolean Operator	Usefulness	Search keywords with Boolean operators
AND	Use the "AND" operator to narrow down the search by requiring both terms to appear in the search results.	“Adaptive Learning” **AND** “Online Platforms” “Adaptive Learning” **AND** “AI Enabled Educational Technologies” “AI-Enabled Adaptive Learning Platforms” **AND** “Learners’ Performance”
OR	Use the "OR" operator to broaden search by including either term in the search results.	“Adaptive Learning” **OR** “Intelligent Tutoring Systems” "online education" **OR** "virtual learning environments" **OR** "e-learning platforms"
NOT	Use the "NOT" operator to exclude specific terms from search results.	"online education" **NOT** "learning management systems"
Parentheses	Use parentheses to group terms the together in order of operations in search query.	**(**"adaptive learning" **OR** "personalized learning"**)** **AND** "artificial intelligence"

The research objective is to trace the impact of an AI-Enabled online adaptive learning platform on learner's performance in terms of learner’s achievement, learning outcome and knowledge accumulated by the learner. It plays a crucial role in guiding the literature survey by providing focus, relevance, gap identification, theoretical framework, evaluation of existing research, and support for hypotheses or research questions.

Research objective also ensures that the literature review is purposeful, systematic, and directly contributes to addressing the research objective of exploring the impact of an AI-Enabled online adaptive learning platform on learner's performance. Extensively the research objective serves as a basis for formulating hypotheses or research questions that align with the specific objective. The literature survey helps in gathering evidence and insights from previous research that can support or inform these hypotheses or research questions, providing a foundation for the empirical investigation.

Research questions serve as a basis for the literature survey in investigating the effectiveness of an AI-Enabled online adaptive learning platform in improving learner's performance. The literature survey will explore the key features and functionalities of such platforms, review existing studies on the impact of adaptive learning on performance, examine the role of AI algorithms in providing personalized feedback, and analyse the relationship between learner engagement, motivation, and performance. It will also consider the influence of adaptive learning strategies, factors affecting effectiveness, challenges in implementation, learner characteristics, contextual variations, and learner perceptions and satisfaction levels. By addressing these research questions through the literature survey, the study aims to gain a comprehensive understanding of the research landscape and inform the design and implementation of AI-Enabled online adaptive learning platforms for improved learner's performance.

For AI-enabled adaptive learning platforms the interrelation between the research objective and research questions could be discussed as follows:

Table 3: Research Objectives & Research Questions

Research Objectives	Research Questions	Key-Relationship
To develop an AI-enabled online adaptive learning platform that can personalize the learning experience for individual learners.	1. How can an AI-enabled online adaptive learning platform be designed to effectively personalize the learning experience for individual learners?	The design and development of the AI-enabled online adaptive learning platform directly contribute to its ability to personalize the learning experience for individual learners.
To investigate the impact of the AI-enabled online adaptive learning platform on learner's performance.	2. What is the impact of the AI-enabled online adaptive learning platform on learner's performance, in terms of academic achievement and knowledge retention?	The investigation of the impact of the AI-enabled online adaptive learning platform on learner's performance helps understand the effectiveness of the platform in improving academic achievement and knowledge retention.
To explore the effectiveness of adaptive learning strategies in enhancing learner engagement and motivation.	3. How do adaptive learning strategies incorporated in the platform affect learner engagement, motivation, and self-efficacy?	The exploration of the effectiveness of adaptive learning strategies contributes to understanding their impact on learner engagement, motivation, and self-efficacy, which in turn can influence learner's performance.
To examine the role of AI algorithms in providing real-time feedback and recommendations to learners.	4. How do the AI algorithms in the platform provide real-time feedback and recommendations to learners, and how do learners respond to this personalized feedback?	The examination of the role of AI algorithms in providing real-time feedback and recommendations helps understand their influence on learner performance and how learners respond to personalized feedback, which can impact their learning outcomes.
To assess the scalability and feasibility of implementing an AI-enabled online adaptive learning platform in various educational settings.	5. What are the challenges and opportunities associated with implementing an AI-enabled online adaptive learning platform in different educational settings, and how can they be addressed?	The assessment of scalability and feasibility of implementing the AI-enabled online adaptive learning platform helps identify challenges and opportunities associated with its implementation, which can have implications for learner's performance in different educational settings.

After reviewing the various research papers the contained research trend has been revealed in AI-Enabled Online Adaptive Learning Platform and how it affects the performance of learners. There are some research trends has been intercepted during the literature review. They are as follows:

- The impact of AI-enabled adaptive learning platforms on academic achievement.
- Effectiveness of personalized recommendations in AI-enabled adaptive learning platforms.
- Learner engagement and its relationship with performance in AI-enabled adaptive learning platforms.
- Adaptive feedback and its impact on learner performance.
- Learner characteristics and their influence on performance in AI-enabled adaptive learning platforms.
- Comparative analysis of different AI algorithms in adaptive learning platforms.
- Longitudinal study on the sustained impact of AI-enabled adaptive learning platforms on learner performance.
- The role of learner self-regulation in AI-enabled adaptive learning platforms and performance outcomes.
- Comparative analysis of learner performance in AI-enabled adaptive learning platforms across different educational contexts.
- Personalized Learning Analytics
- Explainable AI in Adaptive Learning
- Individual Differences and Learner Modelling
- Adaptive Feedback and Interventions
- Ethical Considerations in AI-Enabled Learning
- Transfer of Learning and Real-World Application

3. Interrelated variable matrix for AI-Enabled online adaptive learning platform and learner’s performance

The AI-enabled online adaptive learning platform and learner's performance involves identifying the key variables that can influence and impact learner performance (Zhao, 2023). Those variables could be discussed under the various dimensions.

Variables are as follows in Table 4:

Table 4: Variables for AI-Enabled adaptive learning platforms

Learner Characteristics	• Age • Educational background • Learning style • Motivation level • Prior knowledge
Platform Usage Metrics	• Time spent on the platform • Frequency of platform usage • Number of completed activities • Engagement level (e.g., interaction with content)
Adaptive Learning Algorithm Metrics	• Personalized recommendations provided • Adaptation speed (how quickly the system adapts to user needs) • Accuracy of content alignment with user's proficiency level • Learning path effectiveness
Content Quality and Diversity	• Relevance of the content to the learner's goals • Appropriateness of the difficulty level • Diversity of content formats (e.g., videos, quizzes, interactive exercises) • Currency and accuracy of the information
Learner Performance Metrics	• Achievement of learning objectives • Knowledge retention over time • Improvement in test scores or assessments • Self-reported satisfaction and confidence

More precisely the correlation between variables could be expressed as follows:
Correlation between Learners Characteristics and Platform Uses Matrix

a) Learners with higher motivation levels may spend more time on the platform and engage with activities more frequently.

b) Learners with different learning styles may have varying patterns of platform usage.

Correlation between Platform Usage Metrics and Adaptive Learning Algorithm Metrics

a) Increased platform usage and completion of activities provide more data for the adaptive algorithm to personalize recommendations effectively.

b) Learners who engage more frequently may experience faster adaptation to their needs.

Correlation between Adaptive Learning Algorithm Metrics and Content Quality and Diversity

a) Effective adaptation should result in personalized content recommendations that match the learner's proficiency level and interests.
b) The algorithm's accuracy in aligning content with learner needs contributes to the perception of content quality.

Correlation between Content Quality and Diversity and Learner Performance Metrics

a) High-quality and diverse content that matches the learner's goals and proficiency level can positively impact knowledge acquisition and retention.
b) Relevance and appropriateness of content can enhance satisfaction and confidence, which may lead to improved performance.

The literature survey explores the various interrelated variables. The interrelated variable matrix highlights the relationships between different variables involved in the study of an AI-Enabled online adaptive learning platform and learner's performance. The platform's design, features, and effectiveness of AI algorithms impact learner engagement, motivation, and performance. Adaptive learning strategies incorporated in the platform affect learner's academic achievement, platforms scalability, feasibility, and knowledge retention, which are key components of learner performance.

Table 5: Interrelations between Variables

Variables	Description	Interrelations
AI-Enabled online adaptive learning platform	The platform that incorporates AI algorithms to provide personalized and adaptive learning experiences for learners.	The design and features of the platform influence learner engagement and motivation. The effectiveness of AI algorithms in providing personalized feedback impacts learner performance. Adaptive learning strategies incorporated in the platform affect learner's academic achievement and knowledge retention. The platform's scalability and feasibility in various educational settings influence learner performance. Learner perceptions and satisfaction with the platform can impact their engagement and performance.
Learner performance	The outcome or achievement of learners in terms of academic achievement, knowledge retention, and overall learning effectiveness.	Learner engagement and motivation influence performance outcomes. Personalized feedback and recommendations provided by the platform impact learner performance. Adaptive learning strategies incorporated in the platform affect learner's academic achievement and knowledge retention. Learner characteristics, such as prior knowledge and learning styles, interact with the platform to affect performance outcomes. - The effectiveness of the platform varies across different educational contexts and learner populations, impacting performance outcomes.

Table 5 continued

Learner engagement and motivation	The level of involvement, interest, and drive exhibited by learners towards the learning process and tasks.	The design and features of the platform influence learner engagement and motivation. Adaptive learning strategies incorporated in the platform affect learner engagement and motivation. - Learner engagement and motivation influence performance outcomes.
Personalized feedback and recommendations	AI-generated feedback and recommendations tailored to the individual learner's needs and progress.	Personalized feedback and recommendations provided by the platform impact learner performance. The effectiveness of AI algorithms in providing personalized feedback impacts learner performance.
Adaptive learning strategies	Instructional techniques and approaches that adapt to the learner's progress, preferences, and needs.	Adaptive learning strategies incorporated in the platform affect learner's academic achievement and knowledge retention. Learner engagement and motivation are influenced by adaptive learning strategies.
Platform scalability and feasibility	The ability of the AI-Enabled adaptive learning platform to be implemented and effective in various educational settings.	The platform's scalability and feasibility in various educational settings influence learner performance. The effectiveness of the platform may vary across different educational contexts and learner populations, impacting performance outcomes.
Learner perceptions and satisfaction	Learners' subjective views, experiences, and satisfaction levels regarding the AI-Enabled adaptive learning platform.	Learner perceptions and satisfaction with the platform can impact their engagement and performance. Learner engagement and motivation are influenced by their perceptions and satisfaction with the platform.

An AI-enabled online adaptive learning platform can have several positive effects on a learner's performance. By leveraging advanced technologies and adaptive techniques, these platforms aim to optimize the learning experience, address individual learner unique needs, and ultimately enhance performance outcomes.

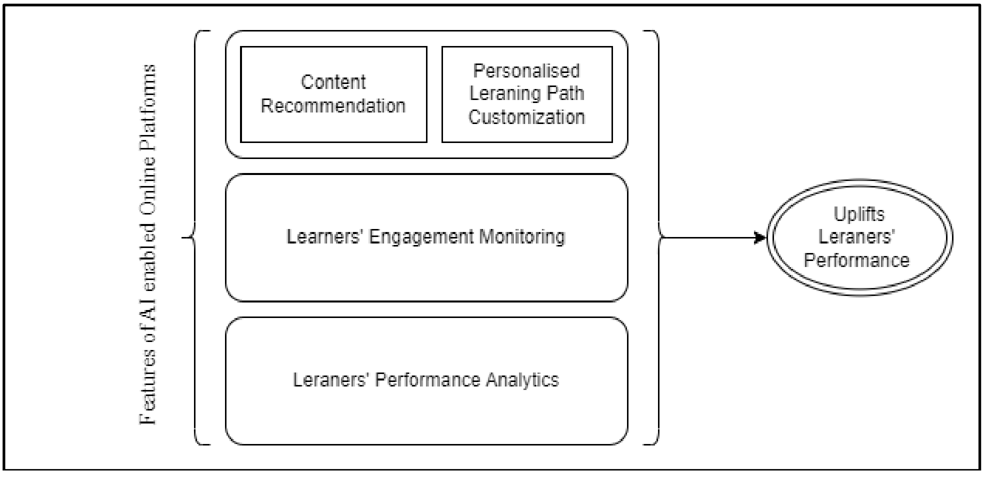
The included features are as follows:

- **Content Recommendation:** This feature represents the platform's ability to recommend appropriate learning materials to learner based on the learner's preferences, learning style and previous performance record (Zhao, 2022).
- **Personalized Learning Path Customization:** This feature expresses the platform's capability to customize the unique learning path for each learner,

fabricate the subject content, pace, or difficulty level according to their individual learning requirements (Reichardt, 2021).

- **Engagement Monitoring:** This indicates the platform's capability of continues monitoring of learner engagement, tracking factors like time spent on assignments or tasks, interaction with learning materials, or active participation in collaborative activities.
- **Performance Analytics:** This feature represents the platform's ability to analyse, intercept and provide feedback on the learner's performance, offering useful insights to learner about strengths, weaknesses, and areas for improvement.

Figure 4: Features of AI enabled Online Platforms



4. Finding of Conducted Literature Review

After the study of various research articles and conference papers, some common themes and findings that have been observed in research related to AI-enabled adaptive learning platforms and learner performance based on various dimensions. The findings could be summarised as follows:

a) **Personalization and Individualized Learning**: AI-enabled adaptive learning platforms can personalize the learning experience for each learner based on their strengths, weaknesses, and learning preferences. Learners receive targeted content and activities to address their specific needs, potentially leading to improved learning outcomes.

b) **Learning Effectiveness:** Several studies have reported positive effects on learner performance when using AI-enabled adaptive learning platforms. Learners tend to show higher engagement, retention, and mastery of learning materials compared to traditional one-size-fits-all approaches.

c) **Real-time Feedback and Support:** AI systems can provide immediate feedback and support to learners. This instant feedback helps learners understand their progress, identify areas for improvement, and adjust their learning strategies accordingly.
d) **Enhanced Engagement:** AI-driven adaptive learning platforms often incorporate interactive elements, gamification, and multimedia content, leading to increased learner engagement and motivation.
e) **Learning Analytics:** AI-enabled platforms can collect and analyse vast amounts of learner data. Learning analytics help educators and administrators identify trends, spot knowledge gaps, and make data-driven decisions to optimize the learning process.
f) **Challenges and Limitations:** Despite the potential benefits, some studies have highlighted challenges related to AI-enabled adaptive learning, such as the need for careful curation of content, data privacy concerns, and potential biases in the algorithms used.
g) **Learner Preferences and Acceptance:** Learners' acceptance and attitudes towards AI-enabled adaptive learning platforms can influence their effectiveness. The studies of Kazoun, (2021) explored factors influencing learner preferences and ways to enhance learner acceptance.
h) **Implementation and Pedagogical Considerations:** Successful integration of AI-enabled adaptive learning requires careful planning, appropriate training for educators, and aligning the technology with pedagogical goals explored by Palanisamy (2021).
i) **Scalability and Accessibility:** AI-enabled adaptive learning platforms offer the potential for scalable and accessible education, providing personalized learning experiences to many learners regardless of their geographical location or learning pace.

4.1. Learners' Performance

The capability of personalization of AI-enabled adaptive learning platform enhances the learners' performances. Kakish (2022) suggested that various features of adaptive learning are responsible to create best learners' performance. Those could be discussed as follows:

Academic Excellence

- The personalised learning path and customised content gives the positive impact to learner through the better academic achievements considered by Peng (2019).

- Approach gives the learning mastery over the traditional teaching methods.
- It addresses the unique learning needs of each individual learner and produce the better learning outcomes.

Engagement and Motivation

- AI-driven adaptive platforms encapsulated interactive learning environment, gamification, and real-time feedback, leading to increased learner engagement.
- Personalized learning experiences create higher motivation and active involvement in the learning process considered by Singh (2023).
- Learners are more engaged, when the platform responds to their specific learning needs and systematically addresses the learning preferences.

Time Efficiency

- Personalized content delivery approach allows learners to maintain the learning progress at their own pace and learner could effectively utilize their time.
- Adaptive learning online platforms adjust the difficulty level of provided content based on learners' proficiency levels and reducing time spent on learning material they already know well.

Knowledge Retention

- Platforms could trace the personalized learning paths based on real time feedback and learners' profile. It is helpful for the learner to improved knowledge retention.
- The adaptive learning approach helps learner for long-term memory.

Individual Differences

- The effectiveness of AI-enabled adaptive learning varies based on individual learner characteristics, such as learning styles, learning pattern, learning performance and prior knowledge about the subject.
- The individual learning paths accommodate diverse learner needs, improving outcomes for various student profiles also suggested by Essa, (2023).

Metacognition and Self-Regulated Learning

- Availability of real-time feedback and analytical progress tracking facilities foster metacognitive awareness of learner and self-regulated learning attitude.
- Learners can monitor their own learning progress and gradual learning outcomes, set learning objectives, and make informed decisions about their study strategies and the selection of appropriate contents suggested by Ghergulescu (2021).

Problem-Solving Skills

- Online adaptive learning platforms with problem-solving approach promote critical thinking and problem-solving skills of learners and help slow learner to be the good learner.
- Learners engage themselves in analytical and creative thinking as they tackle problems of varying difficulty levels.

Learning Autonomy and Independence

- AI-driven online adaptive learning platforms empower learners to take ownership of their learning journey considered by Ghergulescu (2021).
- Learners have more control over their learning pace, content selection, learning outcomes and learning goals.
- Learners gain confidence in their learning abilities due to real-time feedback and personalised support from the adaptive platform intercepted by Chaiyarak (2021).
- Personalized learning experiences help learners develop higher levels of self-efficacy in their learning also assumed by Ezzaim (2022).

4.2. Learners' Engagement

The research papers showing impact of adaptive learning platforms on learner's engagement. Learners' engagement could be discussed under the following themes:

Relevance and Personalised Learning Environment: AI-driven adaptive learning platforms can provide personalized learning experiences customised to each learner's needs, preferences, and proficiency levels. Khosravi (2020) explores that learners are more engaged and associated with learning processes, when the content is relevant to their learning goals and individual capabilities.

Interactive Learning Experience: AI-enabled adaptive platforms often incorporated with interactive learning elements, multimedia content, and simulations, which increase learner engagement and involvement with the environment. Interactive learning activities create the active participation and deeper understanding of the subject material suggested by Zliobaite (2012).

Real-time Feedback and Progress Tracking: Online Adaptive learning platforms provides the immediate feedback on learners' performance and progress. Real-time feedback helps learners stay motivated with the entire process and informed learners about their learning strengths and areas for improvement.

Gamification and Incentives: Gamification and incentives are two powerful approaches that can be used to enhance learning. Gamification is the introduction of game features in non-gaming circumstances while incentives are rewards that are offered to encourage desired learning behaviours. Gamification elements, such as badges, points, and rewards, can be integrated into adaptive learning platforms. These features add an element of fun during learning process and competition, motivating learners to achieve their goals.

Learner Autonomy and Control: AI-enabled adaptive learning platforms strengthen learners by providing them learning autonomy with more control over their learning experience. Alajlani, (2023) presumed that learners can select the appropriate learning paths, set learning goals, and learn at their preferred pace, enhancing their learning engagement and sense of ownership during the learning.

Collaborative Learning Opportunities: Adaptive platforms incorporate collaborative learning features, enabling learners to interact with small groups to learn new concepts, exchange ideas, and work together on projects or complete the project. Collaborative activities can enhance engagement through social interaction and beneficial for the students.

Learning Analytics and Insights: AI-driven platforms accumulate high volume data on learners' interactions and performance. Learning analytics provide valuable insights to educators with the support of various data mining algorithms. Platforms also identify engagement patterns and optimize the learning experience.

Accessibility and Flexibility: The dynamic nature of AI-enabled adaptive platforms offer flexibility in accessing learning materials, allowing learners to continue study anytime and anywhere. The customise accessibility of learning content enhances learner engagement and promotes a self-directed learning approach.

4.3. Technologies for AI-Enabled adaptive learning platforms

It is considered by Kazimzade (2019) that AI-enabled adaptive learning online platforms utilise various modern technologies to develop and deliver personalised and effective learning experiences to learners. These technologies work jointly to analyse data, continues assessment of learner progress, and dynamically adopted content delivery and learning pathways. The useful technologies for the platforms are as given below:-

Machine Learning (ML) & Deep Learning (DL): Machine learning and Deep Learning are at the core of AI-enabled adaptive learning platforms. ML and DL algorithms can analyse procured learner data to identify learning patterns, learning preferences, and learning needs. They use this extracted information to personalise content, recommend resources, and adjust the learning experience in a real-time environment.

Data Analytics: Data analytics techniques help to complete the entire learning process and interpret large amounts of learner data. Learning analytics provide comprehensive insights into learner behaviour, progress, and engagement. It enables educators to make informed decisions and optimise the learning experience.

Natural Language Processing (NLP): NLP makes AI-enabled adaptive learning platforms capable of understanding and generating human language. NLP enables chatbots, virtual assistants, and automated feedback systems, enhancing learner interactions, involvement, and support.

Recommendation Systems: Recommendation systems extensively deploy ML algorithms to suggest relevant learning content, resources, and activities based on learners' preferences, interests, and performance. These systems gradually improve learner engagement and facilitate personalised learning pathways.

Adaptive Assessment: Adaptive assessment technologies use machine learning (ML) and data analytics to tailor assessments based on learners' proficiency levels. These assessments dynamically adjust the difficulty and align the sequence of questions to provide accurate and informative evaluations of learner knowledge also supported by Shuguang, (2021).

Content Curation and Generation: AI-powered content curation systems help select and organise relevant learning materials or contents based on learners' unique needs. AI can also generate the required learning content on a real-time basis, such as personalised quizzes or exercises, to supplement and strengthen instructional resources.

Predictive Analytics: Predictive analytics utilise historical learner data and machine learning (ML) algorithms to forecast probable learners' performance and identify potential future challenges during the process of learning. Educators or subject instructors can intervene early to support struggling learners.

Personalized Learning Pathways: AI-enabled platforms use learner data and advanced computational algorithms to create individualised learning pathways, tailoring content and performing varied learning activities to upgrade learners' learning strengths.

Chatbots and Virtual Assistants: Chatbots and virtual assistants arrange instant learning support and answers to learners' questions. Kaiss (2023) believes that these AI-driven conversational intelligent agents provide immediate assistance and establish communication between learners and the online platform.

Cognitive Modeling: Cognitive modeling technologies analyse the learners' interactions and responses deeply to understand and map their cognitive processes. This understanding helps adapt specific instructional strategies to suit individual learners.

Virtual Reality (VR) and Augmented Reality (AR): VR and AR technologies have the potential to create immersive and interactive learning experiences for adaptive learning environments. These technologies can be used to simulate real-world scenarios for learning, enhance learners' engagement, and promote the process of active learning.

User Interface (UI) Design: User Interface (UI) Design is a user-friendly and intuitive interface that allows learners to access content without any hindrance, view real-time progress, and receive personalised learning recommendations. It increases student engagement, improves understanding, and personalises the learning experience by adjusting the layout of the content, the use of colour and typography, and the overall visual design of the learning environment.

Security and Data Privacy: Adaptive learning platforms collect and use a significant amount of personal data about learners, including their performance data, demographic information, and interaction data. This data can be used to personalise learning experiences, but it also raises data privacy concerns. The Secure Data Handling involves implementing secure data storage facilities, advanced encryption technologies, and user authentication algorithms to protect learner data for AI-enabled adaptive learning platforms. It also ensures compliance with data privacy regulations and obtains informed consent from learners (Meijuan, 2020).

4.4. Available frameworks for AI enabled adaptive learning platforms

Several frameworks and platforms are available for building AI-enabled adaptive learning platforms. These frameworks offer tools, libraries, and APIs to simplify and facilitate the integration of AI technologies into adaptive learning systems. Researchers commonly discuss popular machine learning frameworks like TensorFlow, Keras, PyTorch, and scikit-learn, which are widely used for building AI models in adaptive learning platforms. Literature reviews highlight adaptive learning engines that leverage machine learning algorithms to personalize content delivery and assessment for individual learners (Roy, 2023). Adaptive learning engines use data from learner interactions to dynamically adjust learning pathways and content. Researchers delve into adaptive assessment technologies that dynamically adjust the difficulty and sequence of questions based on learners' proficiency levels. These assessments provide accurate evaluations of learners' knowledge and skills, it is supported by Roy (2023) and Gupta (2023).

Table 6: Type of Framework

Framework	Description	AI Technology	Features and Capabilities
TensorFlow	Open-source machine learning library developed by Google.	Machine Learning (ML)	Provides a flexible ecosystem for building AI models.
Keras	High-level neural networks API written in Python.	Machine Learning (ML)	Simplifies the development of AI models.
PyTorch	Open-source machine learning library developed by Facebook.	Machine Learning (ML)	Known for its dynamic computation capabilities.
Cognii	AI-powered educational technology company offering an assessment and feedback platform.	Natural Language Processing (NLP)	Provides personalized learning support with automated assessments and feedback.
ALEKS	Adaptive learning platform by McGraw-Hill that uses AI for personalized learning paths and content delivery.	Machine Learning (ML)	Assesses learners' knowledge and delivers personalized learning paths.
Smart Sparrow	Adaptive learning platform with courseware authoring tools for personalized learning experiences.	Machine Learning (ML)	Empowers educators to create personalized learning content.
Area9 Rhapsode	Adaptive learning platform using AI to create personalized learning paths and improve learner outcomes.	Machine Learning (ML)	Delivers personalized learning experiences based on learner performance.
D2L Brightspace	Learning management system with adaptive learning features that personalize learning experiences for students.	Machine Learning (ML)	Uses AI to tailor learning experiences based on learner needs.
ALEKS PPL	AI-powered placement, preparation, and learning platform for personalized learning in various subjects.	Machine Learning (ML)	Provides personalized learning paths and content for students.
Cerego	Adaptive learning platform using AI algorithms to optimize content delivery based on learners' performance.	Machine Learning (ML)	Optimizes content delivery based on learner knowledge levels.

4.5. Data analytics algorithms for AI enabled adaptive learning platform

An AI-enabled adaptive learning platform plug-in with various data analytics algorithms to process and interpret procured learner data for providing personalised learning experiences. These data analytics algorithms are helpful in analysing learners' interactions, behaviours, and preferences to tailor content, activities, and assessments according to individual needs. Below is a more in-depth exploration of the data analytics algorithms commonly used on such platforms:

Collaborative Filtering: Collaborative filtering is a widely used recommendation algorithm that focuses on finding similarities between learners to make personalised content suggestions. It analyses the historical interactions and preferences of similar learners to recommend relevant learning materials. The recommender system used this technique to recommend learning content to learners based on the ratings or preferences of other learners (Bajaj, 2018).

User-Based Collaborative Filtering: This approach identifies learners with similar behaviour and preferences and recommends content based on what similar users have found useful.

Item-Based Collaborative Filtering: Instead of grouping learners, this method identifies similar learning resources and suggests items based on their relevance to the learner's past interactions.

Content-Based Filtering: The content-based filtering algorithm suggests learning materials to learners based on the content attributes and the learner's historical data. It leverages natural language processing (NLP) techniques to analyse the characteristics of the content and match it with the learner's learning interests and learning style.

Matrix Factorization: Matrix factorization techniques, such as Singular Value Decomposition (SVD) and Alternating Least Squares (ALS), are used for collaborative filtering. These algorithms decompose the learner-item interaction matrix to discover latent factors that influence the learner's preferences and content relevance. This is used in machine learning to decompose a matrix into two or more matrices of lower rank. This algorithm is applicable for a variety of tasks, including dimensionality reduction, recommendation systems, and collaborative filtering. In adaptive learning, matrix factorization can be used to learn a model of the learner's knowledge state. Additionally, models can be used to personalise the learning experience for the learner by providing them with content that is most relevant to their current level of knowledge.

Clustering Algorithms: Clustering algorithms group learners with similar characteristics or behaviours together. By segmenting learners into clusters, the adaptive learning platform can provide personalised content and activities tailored to each group's preferences and needs. K-means, hierarchical clustering, and density-based clustering are common methods used in adaptive learning platforms.

Decision Trees: Decision trees are employed to create learner profiles and identify relevant content based on their attributes and interactions. These trees can help determine the optimal learning path for individual learners, considering their strengths and weaknesses. Adaptive learning platforms could use decision trees to create a model of the learner's knowledge state. This model can then be used to recommend content that is most relevant to the learner's current level of knowledge. For example, in a mathematics course, the decision tree model could be used to assess the learner's understanding of different mathematical concepts. The model could then be used to recommend problems that are at the right level of difficulty for the learner and to provide feedback that is tailored to their specific needs. Decision trees can also be used to track the learner's progress over time. This information can be used to identify areas where the learner is struggling and provide them with additional support.

Reinforcement Learning: Reinforcement learning algorithms can adapt the learning experience based on the learner's performance and feedback. The platform can adjust the difficulty level of content or introduce challenges based on the learner's progress, helping to maintain an optimal level of engagement.

Natural Language Processing (NLP) Techniques: NLP algorithms are useful to analyse learner feedback, comments, and responses to understand their sentiments and needs. Sentiment analysis, text classification, and named entity recognition are examples of NLP techniques used to gather valuable insights for personalization.

Time Series Analysis: Time series analysis is deployed to understand the temporal patterns in a learner's progress and engagement. It helps identify trends, periodic behaviour, and anomalies, which can be used to optimise the learning experience over time.

Association Rule Mining: Association rule mining identifies patterns and relationships between different learning resources and learner interactions. By discovering associations between activities, assessments, and learner outcomes, the platform can suggest complementary materials or activities.

Bayesian Networks: Bayesian networks model probabilistic relationships between different factors and variables in the adaptive learning environment. They are useful for predicting learner behaviour and outcomes, enabling the platform to proactively adjust content and activities to meet learners' changing needs.

Deep Learning Algorithms: Deep learning models, such as neural networks, can be employed for complex and intricate data analysis tasks. They excel at learning patterns from vast amounts of learner data, enabling more accurate personalization and adaptive learning experiences.

Contextual Bandits: Contextual Bandits algorithms combine exploration and exploitation to optimise personalised content recommendations. They consider both the learner's known preferences and explore new options to continually improve the learning experience. Contextual bandits are a type of online learning algorithm that can be used to make sequential decisions in an environment where the rewards are not immediately known. This makes them a good fit for adaptive learning systems, where the goal is to personalise the learning experience for the learner by providing them with content that is most relevant to their current level of knowledge. In contextual bandits, each decision is made in the context of a particular set of features. These features can be anything that is known about the learner at the time of the decision, such as their past performance, their interests, or the current state of the learning material. The goal of a contextual bandit algorithm is to maximise the expected reward over time. This is done by learning a model of the relationship between the features and the rewards. The model is then used to predict the reward for each possible action in the current context.

Table 7: Algorithm for Adaptive Learning

Algorithm	Description	Application in Adaptive Learning
Collaborative Filtering	Recommends content based on similarities among learners' behavior and preferences.	Personalized content recommendations based on what similar learners have found useful.
Content-Based Filtering	Recommends content based on the attributes and characteristics of the learning materials.	Personalized content recommendations based on learners' interests and learning style.
Matrix Factorization	Decomposes the learner-item interaction matrix to discover latent factors influencing preferences .	Improving collaborative filtering-based recommendations through matrix decomposition.
Clustering	Groups learners with similar characteristics or behaviours into clusters.	Segmenting learners into cohorts for targeted content and activities.
Decision Trees	Creates learner profiles and makes decisions based on attribute values.	Personalizing learning paths and content based on learners' characteristics and progress.
Reinforcement Learning	Adapts the learning experience based on feedback and rewards.	Adjusting content difficulty and challenges based on learners' performance and engagement.
Natural Language Processing	Analyzes text data (learner feedback, comments) to extract insights.	Understanding learner sentiments, identifying areas of improvement, and feedback analysis.
Time Series Analysis	Analyzes temporal patterns in learners' progress and engagement.	Identifying trends, periodic behavior, and anomalies in learner behavior.
Association Rule Mining	Identifies patterns and relationships between different learning resources and interactions.	Suggesting complementary materials or activities based on learner interactions.
Bayesian Networks	Models probabilistic relationships between variables in the learning environment.	Predicting learner behavior and outcomes based on interrelated variables.
Deep Learning Algorithms	Utilizes neural networks to process vast amounts of learner data for complex analyses.	Enhancing data analysis and personalization capabilities for various algorithms.
Contextual Bandits	Combines exploration and exploitation to optimize personalized content recommendations.	Balancing content suggestions based on both known preferences and exploring new options.

The choice of data analytics algorithms depends on the nature of the data, the complexity of the adaptive learning platform, and the specific learning objectives. Often, a combination of these algorithms is used to provide comprehensive and effective personalised learning experiences for users. As the field of AI and data analytics evolves, new algorithms and techniques may emerge to further enhance the capabilities of AI-enabled adaptive learning platforms.

5. Conclusion

The literature review on AI-enabled adaptive learning platforms and learner performance presents compelling evidence of the positive impact of AI technologies on educational outcomes. Through an in-depth analysis of various research studies and academic publications, the following comprehensive conclusion emerges:

AI-enabled adaptive learning platforms have emerged as a transformative force in education, redefining learner's engagement approaches with educational content and optimising their learning experiences and performances. These platforms leverage artificial intelligence and machine learning algorithms to deliver a personalised, data-driven, and adaptive learning environment to individual learners, satisfying their unique needs, learning preferences, and proficiency levels.

The adaptive nature of these online learning platforms, combined with continuous feedback mechanisms and data analysis, has proven to be instrumental in enhancing learner performance and learning outcomes across diverse educational settings and disciplines.

One of the most significant findings from the literature review is that online adaptive learning platforms provide substantial improvements in learner engagement and motivation. Learners respond actively to adaptive content delivery that perfectly aligns with their learning interests, prior knowledge of the subject, and learning styles [8]. The ability of AI-powered online adaptive learning platforms to dynamically adjust learning materials and assessments in real-time based on learner interactions to develop a sense of ownership and control over the learning process, promoting self-directed learning and intrinsic motivation, leads learners to become active participants in their educational journey, leading to increased commitment and perseverance in mastering complex concepts, skills, or knowledge sets.

Furthermore, AI-enabled adaptive learning platforms have been shown to significantly impact learner performance and knowledge retention. Learners benefit from tailored instruction, immediate feedback, and targeted interventions, enabling them to address knowledge gaps or learning gaps effectively and progress at their optimal pace. By continuously adapting the learning experience to challenge and reinforce learners appropriately, adaptive platforms promote deeper understanding and long-term retention of information. The adaptive assessment systems are particularly noteworthy as they ensure that learners encounter questions that match their skill levels, resulting in more accurate and informative evaluations of their knowledge and skills.

Additionally, the literature review emphasises the potential of AI-enabled adaptive learning platforms to address individual differences among learners. By recognising and accommodating diverse learning needs, these platforms create inclusive, interactive, and equitable learning environments. Learners with varying levels of prior knowledge, learning styles, and abilities all benefit from tailored learning experiences that maximise their potential for successful learning outcomes. This aspect is particularly crucial in promoting lifelong learning and supporting learners at different stages of their educational journey.

While the literature review systematically highlights the positive impact of AI-enabled online adaptive learning platforms on learner performance, some limitations and challenges should be considered during the review studies. Privacy concerns and ethical considerations regarding learner data and algorithmic bias require close attention to ensure that learners' information is handled responsibly and equitably. Furthermore, scalability, integration with traditional educational systems, and accessibility issues need to be addressed for widespread adoption and implementation of AI in the modern education system.

In conclusion, the literature review provides evidence that AI-enabled adaptive learning platforms are providing transformative innovation in education. These platforms have the potential to revolutionise the learning experience, empowering learners to achieve higher levels of engagement, motivation, and academic success. The continuous advancements in AI technologies and educational research are likely to further enhance the effectiveness and impact of AI-enabled adaptive learning platforms, preparing the ground for a more personalised and effective education landscape. However, to explore the potential of adaptive learning in education, ongoing research, collaboration between educators and technologists, and thoughtful policy considerations are essential. In conclusion, the literature review provides evidence that AI-enabled adaptive learning platforms are providing transformative innovation in education. These platforms have the potential to revolutionise the learning experience, empowering learners to achieve higher levels of engagement, motivation, and academic success. The continuous advancements in AI technologies and educational research are likely to further enhance the effectiveness and impact of AI-enabled adaptive learning platforms, preparing the ground for a more personalised and effective education landscape. However, to explore the potential of adaptive learning in education, ongoing research, collaboration between educators and technologists, and thoughtful policy considerations are essential. In conclusion, the literature review provides evidence that AI-enabled adaptive learning platforms are providing transformative innovation in education. These platforms have the potential to revolutionise the learning experience, empowering learners to achieve higher levels of engagement, motivation, and academic success. The continuous advancements in AI technologies and educational research are likely to further enhance the effectiveness and impact of AI-enabled adaptive learning platforms, preparing the ground for a more personalised and effective education landscape. However, to explore the potential of adaptive learning in education, ongoing research, collaboration between educators and technologists, and thoughtful policy considerations are essential. In conclusion, the literature review provides evidence that AI-enabled adaptive learning platforms are providing transformative innovation in education. These platforms have the potential to revolutionise the learning experience, empowering learners to achieve higher levels of engagement, motivation, and academic success. The continuous advancements in AI technologies and educational research are likely to further enhance the effectiveness and impact of AI-enabled adaptive learning platforms, preparing the ground for a more personalised and effective education landscape. However, to explore the potential of adaptive learning in education, ongoing research, collaboration between educators and technologists, and thoughtful policy considerations are essential. In conclusion, the literature review provides evidence that AI-enabled adaptive learning platforms are providing transformative innovation in education. These platforms have the potential to revolutionise the learning experience, empowering learners to achieve higher levels of engagement, motivation, and academic success. The continuous advancements in AI technologies and educational research are likely to further enhance the effectiveness and impact of AI-enabled adaptive learning platforms, preparing the ground for a more personalised and effective education landscape. However, to explore the potential of adaptive learning in education, ongoing research, collaboration between educators and technologists, and thoughtful policy considerations are essential.

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